

(1001CJA621120013)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

20-09-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B,C,D	A,B,C	B,C,D	A,B,C,D	A,D	A,C,D	A,B,C	B,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2.00	180.00	8.00	1.54	2.00	0.86 to 0.87	1.23	0.81		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		Q,R,T	Q,R	S	P,Q,R		Q,T	R,S	P	R,S	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,C,D	A,B,C	A,B,D	A,B,D	A,B,C	B,D	A,B,C,D	A,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.00	6.00	3.00	4.00	2.10 to 2.30	1.00	3.00	6.00		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		Q	T	P	T		Q,S	P,R,S	T	Q,T	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A	A,C	C,D	B,C	A,C,D	A,D	B,D	B,C	A,C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	8.00	3.46	3.00	0.00	3.73	4.00	100.00	49.00		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,S,T	Q	R	S		T	Q,S	R	P	

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JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B,C	A,B	B,C,D	B,C	A,C	A,B,C,D	D	D	A
	Q.	11	12								
	A.	A,B,D	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	5.00	1.00	13.33	9.00	1.00	4.00	3.00	0.37 to 0.38		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,D	A,C,D	A,B,C,D	A,B,D	A,C	B,D	B,C	C	A
	Q.	11	12								
	A.	B	D								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2.00	0.25	0.14	4.00	7.00	2.00	5.90 or 5.92	5.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,B,C	A,B	B,C	A,B,C	D	A,B,D	A,B,C,D	A	C
	Q.	11	12								
	A.	B	B								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	3.50	0.50	1.66 or 1.67	1.11	2.00	4.00	9.00	64.00		



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

20-09-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPs)

PAPER-1

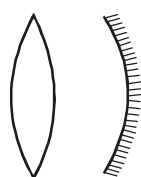
PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A,C)

$$\mu = 3/2$$



$$R = 20\text{cm}$$

Sol.

$$P = 2P_L + P_m$$

$$= -\frac{1}{50} + \frac{1}{10}$$

$$-\frac{1}{f_m} = \frac{4}{50} \Rightarrow f_m = -12.5$$

$$P_L = \frac{3/2 - 5/3}{5/3} \left(\frac{2}{20} \right)$$

$$= -\frac{1}{10} \times \frac{2}{20} = -\frac{1}{100}$$

$$P_m = -\frac{1}{-10} = \frac{1}{10}$$

2. Ans. (A,B,C,D)

Sol. Focal length of upper part $f_1 = 100\text{ cm}$ Focal length of lower part $f_2 = 200\text{ cm}$

$$\frac{1}{v_1} - \frac{1}{-v} = \frac{1}{100}$$

$$\frac{1}{-v_1} - \frac{1}{-v} = \frac{1}{200}$$

$$\Rightarrow \frac{2}{v} = \frac{3}{200}$$

$$\Rightarrow u = \frac{400}{3}$$

3. Ans. (A,B,C)

4. Ans. (B,C,D)

5. Ans. (A,B,C,D)

Sol. $Q_a = +ve$

$$Q_b = -ve$$

$$\frac{KQ_a}{(\ell + x_1)^2} = \frac{KQ_b}{x_1^2}$$

$$\Rightarrow \frac{Q_a}{Q_b} = \left(\frac{\ell + x_1}{x_1} \right)^2 = 64$$

$$E = \frac{KQ_a}{(\ell + z)^2} - \frac{KQ_b}{x^2}$$

$$\frac{dE}{dx} = -2 \frac{KQ_a}{(\ell + x)^3} + \frac{2KQ_b}{x^3} = 0$$

$$\left(\frac{\ell + x}{x} \right)^3 = \frac{Q_a}{Q_b} = \frac{(\ell + x_1)^2}{x_1^2}$$

$$\frac{\ell}{x} + 1 = \frac{(\ell + x_1)^{2/3}}{x_1^{2/3}}$$

$$x = x_2 = \frac{\ell x_1^{2/3}}{(\ell + x_1)^{2/3} - x_1^{2/3}} = 7/3 \text{ m}$$

6. Ans. (A,D)

$$\text{Sol. } \frac{1}{f_{\text{air}}} = \frac{(3/2 - 1)}{1} \left(\frac{2}{R} \right)$$

$$\frac{1}{f_{\text{air}}} = \frac{1}{R}$$

$$f_{\text{water}} = 4 f_{\text{air}}$$

$$P_{\text{water}} = \frac{1}{4R} = \frac{1}{4f_{\text{air}}} = \frac{P_{\text{air}}}{4}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\text{air}}} \quad \left| \frac{v}{u} \right| = 3$$

$$\frac{1}{3U} - \frac{1}{-U} = \frac{1}{f_{\text{air}}}$$

$$\frac{4}{3U} - \frac{1}{f_{\text{air}}} \Rightarrow U = \frac{4f_{\text{air}}}{3} = \frac{f\omega}{3}$$

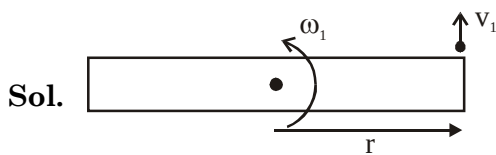
$$\frac{1}{v} - \frac{1}{-\frac{f_w}{3}} = \frac{1}{f\omega}$$

$$\frac{1}{v} + \frac{3}{f_w} = \frac{1}{f\omega}$$

$$\frac{1}{v} = -\frac{2}{f\omega}$$

$$v = \frac{-f_w}{2} \text{ virtual image}$$

7. Ans. (A,C,D)



$$I = \frac{m4r^2}{12} = \frac{mr^2}{3}$$

$$L_i = \frac{mr^2}{3} \omega - mr \frac{\omega r}{4} = \frac{mr^2 \omega}{12}$$

$$L_f = \frac{mr^2}{3} \omega_1 = mrv_1$$

$$L_i = L_f \Rightarrow \frac{mr^2 \omega}{12} = \frac{mr^2}{3} \omega_1 + mrv_1$$

$$\frac{r\omega}{12} = \frac{r\omega_1}{3} + v_1 \quad \dots(i)$$

$$e = \frac{1}{2} = \frac{-r\omega_1 + v_1}{r\omega + \frac{r\omega}{4}}$$

$$\Rightarrow v_1 - r\omega_1 = \frac{5r\omega}{8} \quad \dots(ii)$$

$$3 \times (i) + (ii) \Rightarrow 4v_1 = \frac{5r\omega}{8} + \frac{r\omega}{4} = \frac{7r\omega}{8}$$

$$v_1 = \frac{7r\omega}{32}$$

$$\omega_1 = -\frac{13r\omega}{32}$$

$$J = P_f - P_i = mrv_1 + \frac{mr\omega}{4} = \frac{15r\omega}{32}$$

8. Ans. (A,B,C)

Sol. $\omega = \frac{v}{r}$

$$v_{CM} = \frac{r}{\sqrt{2}} \omega = \frac{v}{\sqrt{2}}$$

$$I_0 = \frac{m(\sqrt{2}r)^2}{12} + m\left(\frac{r}{\sqrt{2}}\right)^2$$

$$L = I_0 \omega$$

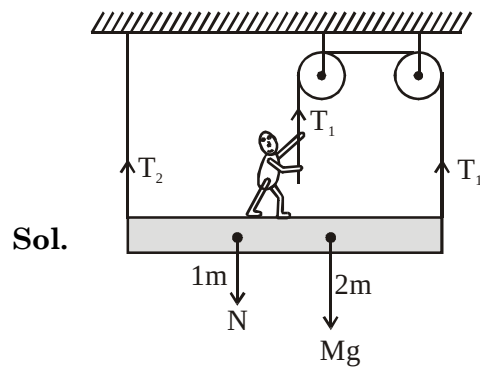
$$= \frac{mr^2}{6} + \frac{mr^2}{2} = \frac{2mr^2}{3}$$

$$\frac{2Mrv}{3}$$

$$\text{Translational KE} = \frac{1}{2} mv_{CM}^2 = \frac{mv^2}{4}$$

$$\text{Rotational KE} = \frac{1}{2} I_{CM} \omega^2 = \frac{1}{2} \frac{mr^2}{6} \frac{v^2}{r^2} = \frac{mv^2}{12}$$

9. Ans. (B,C,D)



$$T_1 + N = Mg \quad \dots(i)$$

$$T_1 + T_2 = M + Mg \quad \dots(ii)$$

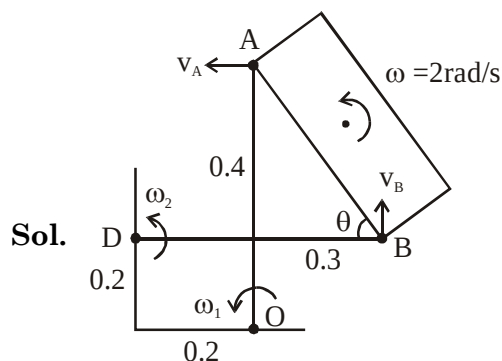
$$N \times 1 + Mg \frac{3}{2} = T_1 \times 3 \quad \dots(iii)$$

$$\frac{5Mg}{2} = 4T_1$$

$$T_1 = \frac{5Mg}{8} \quad N = \frac{3Mg}{8}$$

$$T_2 = \frac{3Mg}{4}$$

10. Ans. (A,B,D)



$$\tan \theta = \frac{4}{3}$$

$$\theta = 53^\circ$$

$$v_B = 0.5\omega_2$$

$$v_A = 0.6\omega_1$$

along AB

$$v_A \cos \theta = v_B \sin \theta$$

$$v_A \frac{3}{5} = v_B \frac{4}{5}$$

$$3v_A = 4v_B$$

$$\omega = \frac{v_A \sin \theta + v_B \cos \theta}{0.5}$$

$$= \frac{v_A \frac{4}{5} + \frac{3v_A}{4} \times \frac{3}{5}}{0.5} = 2$$

$$\left(\frac{4}{5} + \frac{9}{20} \right) v_A = 1$$

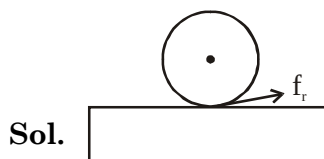
$$v_A = \frac{20}{25} = \frac{4}{5} \text{ m/s}$$

$$v_B = \frac{3}{5} \text{ m/s}$$

$$\omega_1 = \frac{4}{3}; \omega_2 = \frac{6}{5}$$

SECTION-II

1. Ans. 2.00



$$\int r f_r dt = \frac{2}{5} m r^2 \omega$$

$$\int f_r dt = 2m = m v_{cm}$$

$$v_{cm} = 2 \text{ m/s}$$

2. Ans. 180.00

3. Ans. 8.00

Sol.

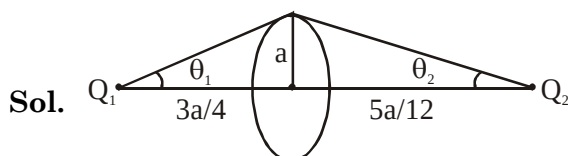
$$\frac{1.5}{v} - \frac{1}{u} = \frac{1.5 - 1}{R}$$

$$\frac{1.5}{v} + \frac{1}{6} = \frac{0.5}{1} \Rightarrow v = 4.5$$

$$v_I = \frac{v^2}{u^2} \frac{v_0}{\mu} = \frac{(4.5)^2}{6^2} \frac{v_0}{1.5} = 3$$

$$v_0 = \frac{3 \times 3}{2} \times \frac{36 \times 4}{9 \times 9} = 8 \text{ m/s}$$

4. Ans. 1.54



$$\tan \theta_1 = \frac{4}{3}$$

$$\tan \theta_2 = \frac{12}{5}$$

$$\frac{\theta_1}{\epsilon_0} \frac{2\pi(1 - \cos \theta_1)}{4\pi} = \frac{\theta_2}{\epsilon_0} \frac{2\pi(1 - \cos \theta_2)}{4\pi}$$

$$\frac{\theta_1}{\theta_2} = \frac{1 - \cos \theta_2}{1 - \cos \theta_1} = \frac{1 - 5/13}{1 - 3/5}$$

$$= \frac{8 \times 5}{13 \times 2} = \frac{20}{12} = 1.54$$

5. Ans. 2.00

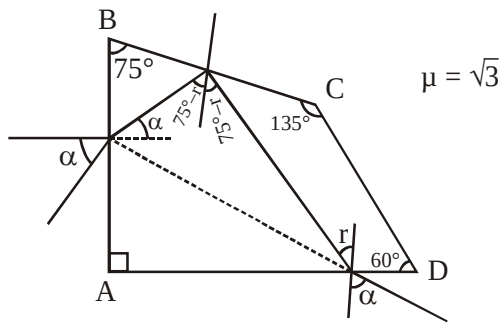
Sol.

$$\mu mg \cos \theta R = \frac{m R^2}{2} \alpha$$

$$\alpha = \frac{2\mu g \cos \theta}{R} = \frac{10}{t}$$

$$t = \frac{10 \times 2}{2 \times \frac{1}{2} \times 10} = 2 \text{ sec}$$

6. Ans. 0.86 to 0.87



Sol.

$$135^\circ + 60^\circ + 90^\circ + r + 15^\circ + r = 360$$

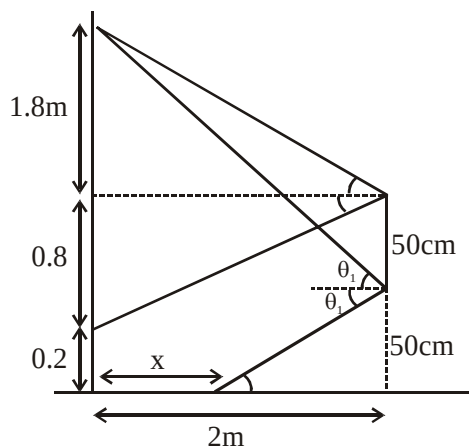
$$2r = 60^\circ$$

$$r = 30^\circ$$

$$\sin \alpha = \mu \sin r$$

$$\sin \alpha = 0.87$$

7. Ans. 1.23



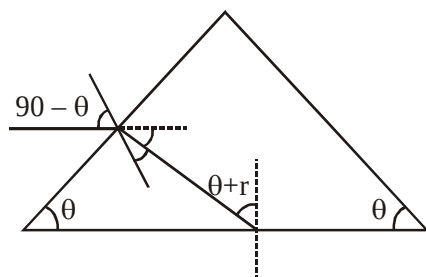
Sol.

$$\tan \theta_1 = \frac{1.8 - 0.5}{2} = \frac{0.5}{2 - x}$$

$$2 - x = \frac{1}{1.3}$$

$$x = 2 - \frac{1}{1.3} = 1.23$$

8. Ans. 0.81



Sol.

$$90 - \theta - r$$

$$\frac{3}{2} \sin(\theta + r) = \frac{4}{3}$$

$$\sin \theta \cos r + \cos \theta \sin r = \frac{8}{9}$$

$$\sin \theta \cos r + \cos^2 \theta \frac{2}{3}$$

$$\sin^2(1 - \frac{4}{9} \cos^2 \theta) = \left(\frac{8}{9} - \frac{2}{3} \cos^2 \theta\right)^2$$

$$\cos^2 \theta = x$$

$$(1 - x) \left(1 - \frac{4x}{9}\right) = \left(\frac{8}{9} - \frac{2}{3}x\right)^2$$

$$1 - x - \frac{4x}{9} + \frac{4x^2}{9} = \frac{64}{81} + \frac{4}{9}x^2 - \frac{32}{27}x$$

$$\frac{17}{81} = \frac{7}{27}x$$

$$x = \frac{17}{21} = 0.8$$

SECTION-IV

1. Ans. (A) Q,R,T; (B) Q,R; (C) S; (D) P,Q,R

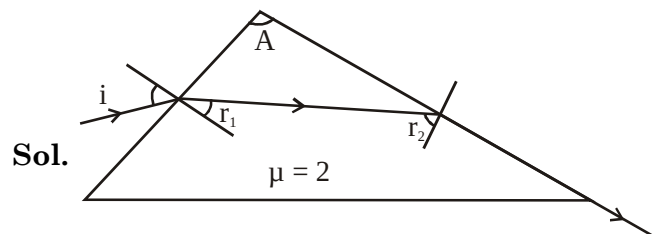
$$\text{Sol. } \mathbf{v}_f = 3\hat{i} + 4\hat{j} + \frac{q}{m}(-100\hat{i} + 50\hat{j})t$$

$$= \left(3 - \frac{100q}{m}t\right)\hat{i} + \left(4 + \frac{50q}{m}t\right)\hat{j}$$

$$\text{speed} = \sqrt{\left(3 - \frac{100qt}{m}\right)^2 + \left(4 + \frac{50qt}{m}\right)^2}$$

$$\text{speed will be minimum at } t = \frac{m}{15q}$$

2. Ans. (A) Q,T; (B) R,S; (C) P; (D) R,S



Sol.

If all rays occur TIR then

$$i = 90^\circ$$

$$r_1 = r_2 = r = A/2$$

$$r_2 > \theta_c$$

$$A/2 > \theta_c$$

$$\sin\left(\frac{A}{2}\right) > \frac{1}{2}$$

$$A > 60^\circ$$

If all rays get transmitted

$$i = 0 \quad r_1 = 0$$

$$r_2 < \theta_c$$

$$A < \theta_c$$

$$\sin A < \sin \theta_c$$

$$\sin A < 1/2$$

$$A < 30^\circ$$

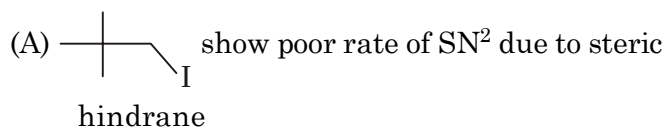
If $30^\circ < A < 60^\circ$ some rays get refracted & some rays get TIR from 2nd surface.

PART-2 : CHEMISTRY

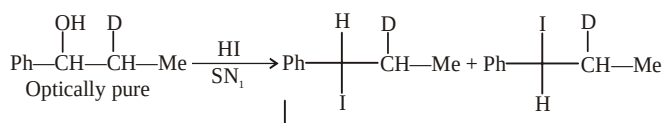
SOLUTION

SECTION-I

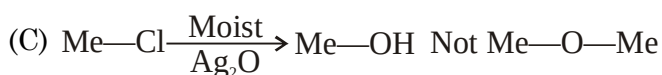
1. Ans.(B,C,D)
2. Ans.(A,C,D)
3. Ans.(A,B,C)
4. Ans.(A, B, D)
5. Ans.(A,B,D)
6. Ans.(A,B,C)
7. Ans. (B,D)
8. Ans. (A,B,C,D)



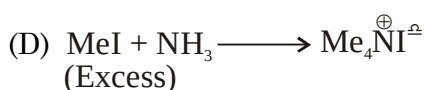
(B)



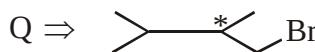
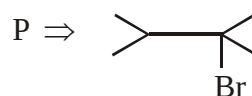
Diastereomer Not Racemic mixture



(It is obtained in dry Ag₂O)



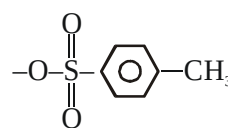
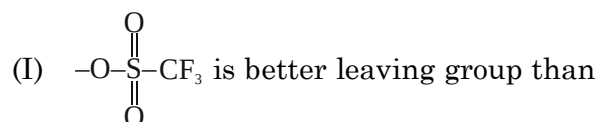
9. Ans (A,C)



10. Ans. (A,C,D)

SECTION-II

1. Ans.(3.00)
2. Ans.(6.00)
3. Ans.(3.00)
4. Ans.(4.00)



(IV) -I is better leaving group than - Br

5. Ans.(2.10 to 2.30)
6. Ans.(1.00)
7. Ans.(3.00)
8. Ans.(6.00)

SECTION-IV

1. Ans (A) $\rightarrow$ (Q); (B) $\rightarrow$ (T); (C) $\rightarrow$ (P); (D) $\rightarrow$ (T)
2. Ans. (A) $\rightarrow$ (Q,S); (B) $\rightarrow$ (P, R, S); (C) $\rightarrow$ (T); (D) $\rightarrow$ (Q, T)

PART-3 : MATHEMATICS**SOLUTION****SECTION-I****1. Ans. (B,C,D)**

$$\text{Sol. } y^2 - 5y + \frac{25}{4} - \frac{25}{4} + 3x + k = 0$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 = -3x + \frac{25}{4} - k$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 = -3\left(x + \frac{k}{3} - \frac{25}{12}\right)$$

(B) Latus Rectum is independent of change of origin.

$$y = 2x + 1 \text{ is tangent}$$

$$(2x + 1)^2 - 5(2x + 1) + 3x + k = 0$$

$$\Rightarrow 4x^2 - 6x + 3x + 1 - 5 + k = 0$$

$$\Rightarrow 4x^2 - 3x + k - 4 = 0$$

$$D = 0$$

$$\Rightarrow 9 - 4(4)(k - 4) = 0 \Rightarrow 9 - 16(k - 4) = 0$$

$$\Rightarrow 16k = 9 + 64 \Rightarrow k = \frac{73}{16}$$

(D) If slope = 0

$$\text{Equation of normal is } y = \frac{5}{2}$$

2. Ans. (A)

$$\text{Sol. Let } P \equiv (2\sqrt{2} \cos \theta, 2 \sin \theta)$$

$$Q \equiv (-2\sqrt{2} \sin \theta, 2 \cos \theta)$$

$$(PQ)^2 = 8(\cos \theta + \sin \theta)^2 + 4(\sin \theta - \cos \theta)^2 = 12 + 4 \sin 2\theta \leq 16$$

$$(PQ)_{\max} = 4$$

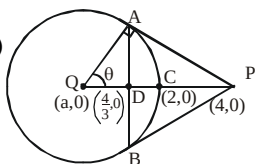
3. Ans. (A,C)

$$\text{Sol. In } \triangle AQD, AQ = AC = 2 - a$$

$$\cos \theta = \frac{QD}{AD} = \frac{\frac{4}{3} - a}{2 - a} \dots (i)$$

in $\triangle PAQ$

$$\cos \theta = \frac{AQ}{QP} = \frac{2 - a}{4 - a} \dots (ii)$$



from (i) and (ii)

$$\frac{\frac{4}{3} - a}{2 - a} = \frac{2 - a}{4 - a} \Rightarrow a = 1 \Rightarrow Q(1,0)$$

$\Rightarrow$ radius $r = 1$

$$\frac{\text{area of } \triangle PAB}{\text{area of } \triangle AQB} = \frac{PQ}{QD} = \frac{4 - \frac{4}{3}}{\frac{4}{3} - 1} = 8$$

4. Ans. (C,D)

Sol. S_2 & S_3 touches each other externally so their common point intersect the join of B & C in $r_2 : r_3$ (i.e. 1 : 2)

$$\Rightarrow (3, -1)$$

Now let the equation of tangent passing through $(3, -1)$ be $y + 1 = m(x - 3) \dots (1)$

if line $\dots (1)$ touches the circle S_1 , then length of perpendicular from centre = Radius

Now do yourself.

5. Ans. (B,C)

Sol. digits are 0, 1, 2, 3, 4, 5

$$(C) \overline{5}.\overline{5}.\overline{4}.\overline{3} = 300$$

$$\Rightarrow \text{Total 4 digit numbers } n = 300$$

$\Rightarrow$ A is incorrect.

(B) Atleast one of the digits must be 0 or 5

$\Rightarrow$ Required numbers = Total - 0 and 5 is not used

$$= 300 - 24 = 276 \Rightarrow D \text{ is incorrect}$$

6. Ans. (A,C,D)

$$\text{Sol. } (1 + x)^{100} = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_{100}x^{100}$$

and differentiation $100(1 + x)^{99}$

$$= C_1 + 2C_2x + 3C_3x^2 + \dots + 100C_{100}x^{99} \dots (i)$$

$$(x + 1)^{100} = C_0x^{100} + C_1x^{99} + C_2x^{98} + \dots + C_{100}x^0 \dots (ii)$$

$$\begin{aligned} \text{in(i)} \times \text{(ii) coeff of } x^{99} & \text{ is } 100 {}^{199}C_{100} = C_1^2 x^{99} \\ & + 2C_2^2 x^{99} + \dots + 100C_{100}^2 \\ 100 {}^{199}C_{100} & = 100 \cdot {}^{199}C_{99} \\ & = 100 \times \frac{199!}{99!100!} = \frac{100}{200} \times \frac{200!}{100!100!} \times 100 = 50 {}^{200}C_{100} \end{aligned}$$

$$\begin{aligned} 50 {}^{200}C_{100} & = 50 \cdot \frac{1.2.3.4. \dots 200}{100! \times 100!} \\ & = 50 \frac{[1.3.5. \dots 199]}{100!} 2^{100} \frac{(1.2.3. \dots 100)}{100!} \\ & = \frac{(1.3.5. \dots 199)}{99!} 2^{99} \end{aligned}$$

7. Ans. (A,D)

Sol. $\sec x + \operatorname{cosec} x = \sin x + \cos x$

$$\begin{aligned} \Rightarrow \frac{\sin x + \cos x}{\sin x \cos x} & = \sin x + \cos x \\ \Rightarrow (\sin x + \cos x)(1 - \sin x \cos x) & = 0 \\ \Rightarrow \sin x = -\cos x \text{ or } 1 & = \sin x \cos x \\ \Rightarrow \tan x = -1 \text{ or } \sin 2x & = 2 \text{ not possible} \\ \Rightarrow x = n\pi - \frac{\pi}{4}, x \in I \end{aligned}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

8. Ans. (B,D)

Sol. For No solution $\Delta = 0$ & unique solution $\Delta \neq 0$

$$\Delta = \begin{vmatrix} 1 & -1 & 3 \\ 2 & -1 & 1 \\ 1 & -2 & \alpha \end{vmatrix} = -\alpha + 2 + 2\alpha - 1 - 12 + 3 = \alpha - 8$$

9. Ans. (B,C)

Sol. $b = \frac{a+c}{2}$

$$c^2 = bd \Rightarrow c^2 = \frac{(a+c)}{2}d \Rightarrow d = \frac{2c^2}{(a+c)}$$

$$d = \frac{2ce}{(c+e)} \Rightarrow \frac{2c^2}{(a+c)} = \frac{2ce}{(c+e)}$$

$$c(c+e) = e(a+c)$$

$$c^2 = ea$$

$$\frac{c}{a} = \frac{e}{c}$$

$$\text{Use C \& D} \Rightarrow \frac{c+a}{c-a} = \frac{e+c}{e-c}$$

10. Ans. (A,C)

AAIOCEAL

(A) -----

IOCL, A,A,A,E

$$\text{required number of words} = \frac{5!}{3!}$$

(B) Required number of words = Total - 'IOCL' is at last four places.

$$= \frac{8!}{3!} - \frac{4!}{3!} = \frac{8!-4!}{3!}$$

$\Rightarrow$ (B) is incorrect.

(C) --- (- - - - -) for A,A,A

$\Rightarrow$ three A are filled at last five places in ${}^5C_3 \cdot 1$ ways

Now, remaining five places are filled by I,O,C,L,E in $5!$ ways.

$\Rightarrow$ required number of words = ${}^5C_3 \cdot 1 \cdot 5!$

(D) A/A/A/E/ ${}^5C_4 \cdot 4! \cdot \frac{4!}{3!}$

(gap method) $\Rightarrow$ D is incorrect.

SECTION-II

1. Ans. 8.00

Sol. Let $\theta = \frac{\pi}{6}$ be the angle opposite to the side

c of $\triangle ABC$, then

$$c^2 = a^2 + b^2 - 2ab \cos \theta \dots\dots\dots(i)$$

$$c^2 = (a-b)^2 + 2ab(1 - \cos \theta)$$

$$= (a-b)^2 + 2ab \left(1 - \frac{\sqrt{3}}{2}\right)$$

$$\text{Also } \Delta = \frac{1}{2}ab \sin \theta \Rightarrow \Delta = \frac{1}{4}ab \Rightarrow ab = 4\Delta$$

$$\text{but } \Delta = 4 \Rightarrow ab = 16 \dots\dots\dots(ii)$$

$$\therefore c^2 = (a-b)^2 + 16 \left(1 - \frac{\sqrt{3}}{2}\right)$$

$\Rightarrow$ c is minimum when $a = b$

i.e. triangle is isosceles

from (ii) $a^2 = 16$

$$\Rightarrow a = 4 \text{ \& } b = 4 \Rightarrow a + b = 8$$

2. **Ans. 3.46**

Sol. Given $\frac{r_1}{r_2} = \frac{2}{1} = \frac{a+x-y}{2a-x-y}$

$$\left(\text{Using } r = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2}\right)$$

$$4a - 2x - 2y = a + x - y$$

$$3x + y = 3a$$

$$y = 3a - 3x$$

$$\text{Now } \frac{BD}{CD} + \frac{AD}{BD}$$

$$\Rightarrow \frac{x}{a-x} + \frac{3(a-x)}{x} \geq 2\sqrt{\left(\frac{x}{a-x}\right) \times \frac{3(a-x)}{x}}$$

$$= 2\sqrt{3}$$

3. **Ans. 3.00**

Sol. k lines parallel & 8 - k lines not parallel

If $k \geq 3$ then

$$\text{Total triangle} = {}^8C_3 - {}^kC_2 \cdot {}^{8-k}C_1 - {}^kC_3$$

$$40 = 56 - \frac{k(k-1)(8-k)}{2} - \frac{k(k-1)(k-2)}{6}$$

$$\Rightarrow k(k-1)(24-3k+k-2) = 96 \Rightarrow$$

$$(k^2 - k)(22 - 2k) = 96$$

$$\Rightarrow 2k^3 - 24k^2 + 22k + 96 = 0 \Rightarrow 2(k-3)$$

$$(k^2 - 9k - 16) = 0$$

$$\Rightarrow k = 3 \because \text{other two values is not integers}$$

$$\text{If } k = 0 \text{ then total triangle } {}^8C_3 = 56$$

$$\text{if } k = 1 \text{ then total triangle } {}^8C_3 = 56$$

$$\text{if } k = 2 \text{ then total triangle } {}^8C_3 - {}^6C_1 = 50$$

4. **Ans. 0.00**

Sol. $[a]x + [a-1]y = 1$

$$x + y = 1 \quad \text{is consistent}$$

$$[a-1]x - [a-2]y = 1$$

$$\Rightarrow \begin{vmatrix} [a] & [a-1] & -1 \\ 1 & 1 & -1 \\ [a-1] & -[a-2] & -1 \end{vmatrix} = 0$$

$$\text{Let } [a] = t \Rightarrow \begin{vmatrix} t & t-1 & 1 \\ 1 & 1 & 1 \\ t-1 & -t+2 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} t-1 & t-2 & 0 \\ 2-t & t-1 & 0 \\ t-1 & -t+2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow (t-1)^2 + (t-2)^2 = 0$$

$$\Rightarrow \text{No value of } t \Rightarrow \text{No value of } a$$

5. **Ans. 3.73**

Sol. $\sin^2 x + \cot^2 x = 3$

$$\text{Let } s = \sin^2 x$$

$$\Rightarrow s + \frac{1-s}{s} = 3 \Rightarrow s^2 + 1 - s = 3s$$

$$\Rightarrow s^2 - 4s + 1 = 0 \Rightarrow s = \frac{4 \pm 2\sqrt{3}}{2}$$

$$\Rightarrow \sin^2 x = 2 - \sqrt{3} \Rightarrow \operatorname{cosec}^2 x = 2 + \sqrt{3}$$

6. **Ans. 4.00**

Sol. $f_n(\theta) = \sum_{r=1}^n \frac{\cos 2r\theta}{\sin(2r+1)\theta \cdot \sin(2r-1)\theta}$

$$= \frac{1}{2\sin\theta} \sum_{r=1}^n \frac{2\cos 2r\theta \sin\theta}{\sin(2r+1)\theta \cdot \sin(2r-1)\theta}$$

$$= \frac{1}{2\sin\theta} \sum_{r=1}^n \frac{\sin(2r+1)\theta - \sin(2r-1)\theta}{\sin(2r+1)\theta \cdot \sin(2r-1)\theta}$$

$$= \frac{1}{2\sin\theta} \left[\sum_{r=1}^n \frac{1}{\sin(2r-1)\theta} - \frac{1}{\sin(2r+1)\theta} \right]$$

$$= \frac{1}{2\sin\theta} \left[\frac{1}{\sin\theta} - \frac{1}{\sin(2n+1)\theta} \right]$$

$$\theta = \frac{\pi}{6}, n = 29$$

$$f_{29}\left(\frac{\pi}{6}\right) = \frac{1}{2 \cdot \sin \frac{\pi}{6}} \left[\frac{1}{\sin \frac{\pi}{6}} - \frac{1}{\sin \frac{59\pi}{6}} \right] = 4$$

7. **Ans. 100.00**

Sol. $x^{\log_{10} y} = 2 \quad \dots(i)$

$$xy = 20 \quad \dots(ii)$$

$$\Rightarrow \text{take } \log_{10} \text{ in equation (ii)}$$

$$\log_{10} x + \log_{10} y = (1 + \log_{10} 2)$$

$$\log_{10} x + \frac{\log_{10} 2}{\log_{10} x} = (1 + \log_{10} 2)$$

$$(\log_{10} x)^2 - (1 + \log_{10} 2) \log_{10} x + \log_{10} 2 = 0$$

$$\Rightarrow \log_{10} x = 1, \log_{10} 2$$

$$\Rightarrow x = 10 \quad \& \quad x = 2$$

$$\Downarrow$$

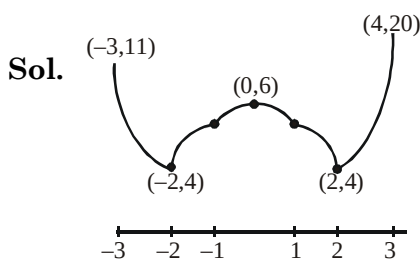
$$y = 2$$

$$y = 10$$

$$\Sigma x = 12, \Sigma y = 12$$

$$100 \cdot \left(\frac{\Sigma x}{\Sigma y} \right) = 100$$

8. Ans. 49.00



For two solutions $k = (6, 11] \cup \{4\}$

$\therefore$ sum of all values = 49

SECTION-IV

1. Ans. (A) $\rightarrow$ (P, S, T); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (R); (D) $\rightarrow$ (S)

Sol. (A) Mid point of (a, b) & (3, 8) lies on $x + 3y = 7$

$$\Rightarrow \frac{a+3}{2} + \frac{3(b+8)}{2} = 7$$

$$\Rightarrow a + 3b + 27 = 14$$

$$\Rightarrow a + 3b + 13 = 0$$

$$\text{Also } m_1 m_2 = -1$$

$$\Rightarrow m_1 m_2 = -1$$

$$\Rightarrow \left(\frac{b-8}{a-3} \right) \left(-\frac{1}{3} \right) = -1$$

$$\Rightarrow 3a = b + 1$$

$$\Rightarrow a = -1 \& b = -4$$

$$\Rightarrow a - b = 3$$

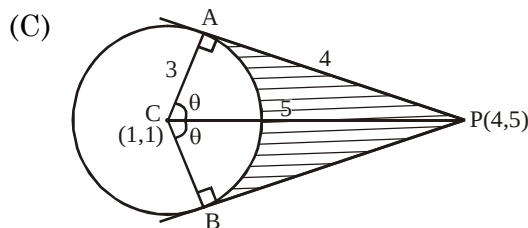
$$(B) \Delta = \frac{1}{2} \begin{vmatrix} \ell & \ell & 1 \\ \ell & 2\ell & 1 \\ 2\ell & \ell & 1 \end{vmatrix}$$

$$= \frac{1}{2} (\ell^2 + \ell^2 - 3\ell^2) = \frac{\ell^2}{2} = 18$$

$$\Rightarrow \ell = \pm 6$$

$$\Rightarrow \text{centroid is } \left(\frac{4\ell}{3}, \frac{4\ell}{3} \right)$$

$$\Rightarrow \frac{|a| + |b|}{4} = 4$$



Shaded area = ar(PCAB) - ar sector(ABC)

$$= 4 \times 3 - \frac{1}{2} \times 9 \times 2\theta$$

$$= 12 - 9\theta$$

$$= 12 - 9 \tan^{-1} \frac{4}{3}$$

$$\Rightarrow \frac{k}{2} = 6$$

$$(D) x^2 y + x^2 - y^3 + 3y^2 - 4 = 0$$

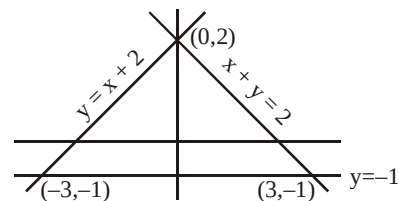
$$x^2 y + x^2 - y^3 - y^2 + 4y^2 - 4 = 0$$

$$x^2(y+1) - y^2(y+1) + 4(y+1)(y-1) = 0$$

$$\Rightarrow (x^2 - y^2 + 4y - 4)(y+1) = 0$$

$$\Rightarrow x^2 = y^2 - 4y + 4 \text{ or } y = -1$$

$$x = y - 2 \text{ or } y = -1$$



$$A = \frac{1}{2} \times 3 \times 6$$

$$A = 9 \text{ sq. units}$$

2. Ans. (A) $\rightarrow$ (T); (B) $\rightarrow$ (Q, S); (C) $\rightarrow$ (R); (D) $\rightarrow$ (P)

$$(A) T_p = aR^{p-1} = 8$$

$$T_q = aR^{q-1} = 12$$

$$T_r = aR^{r-1} = 27$$

$$\Rightarrow 2^{3q-3r+2r-2p} 3^{r-p+3p-3q} = 2^0 \cdot 3^0$$

$$\Rightarrow 2^{3q-r-2p} \cdot 3^{2p+r-3q} = 2^0 \cdot 3^0$$

which is possible when

$$2p + r - 3q = 0$$

infinite solutions

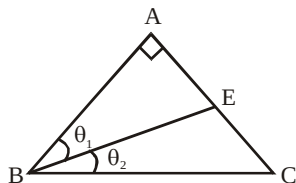
$$(B) \quad L = \frac{b^2 \cos 2A - a^2 \cos 2B}{b^2 - a^2}$$

$$= \frac{(b^2 - a^2) - 2(b^2 \sin^2 A - a^2 \sin^2 B)}{(b^2 - a^2)}$$

$$\therefore b \sin A = a \sin B$$

$$\therefore L = 1$$

(C) From ADE



$$\tan \theta_1 = \frac{AE}{AB} = \frac{AE}{AC} = \frac{1}{2}$$

$$\text{Now } \tan \theta_2 = \tan \left(\frac{\pi}{4} - \theta_1 \right)$$

$$\frac{1 - \tan \theta_1}{1 + \tan \theta_1} = \frac{1 - \frac{1}{2}}{1 + \frac{1}{2}} = \frac{1}{3} \Rightarrow \frac{\tan \theta_1}{\tan \theta_2} = \frac{3}{2}$$

$$(D) \quad x^2 + 9 = 6x \cos x$$

$$\frac{\left(\frac{x}{3} + \frac{3}{x} \right)}{2} = \cos x$$

$$\text{Now } L.H.S \geq 1 \text{ or } \leq -1$$

$$\text{But } -1 \leq R.H.S. \leq 1$$

$$\Rightarrow \text{Either } L.H.S. = R.H.S. = 1$$

which is not possible

$$\therefore L.H.S. = 1 \text{ at } x = 3$$

$$\& R.H.S. \neq 1 \text{ at } x = 3$$

$$\text{OR } L.H.S. = R.H.S. = -1$$

which is again not possible

$$\therefore L.H.S. = -1 \text{ at } x = -3$$

$$\& R.H.S. \neq -1 \text{ at } x = -3$$

$$\Rightarrow \text{No solution}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

20-09-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A,C)

Sol. When parallel light falls from left side

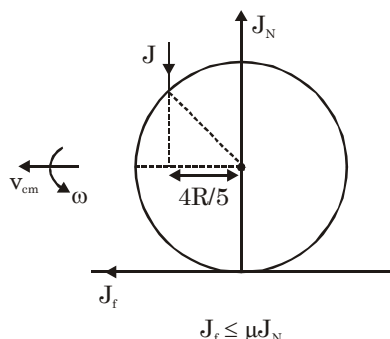
$$\frac{\mu_3}{f} = \frac{(\mu_3 - \mu_2)}{R} + \frac{(\mu_2 - \mu_1)}{R}$$

$$\frac{4/3}{f_1} = \frac{(4/3 - 2)}{R} + \frac{(2 - 1)}{R} \Rightarrow f_1 = 80\text{cm}$$

When parallel light falls from right side

$$\frac{1}{f_0} = \frac{(1-2)}{-R} + \frac{(2-4/3)}{-R} \Rightarrow f_2 = 60\text{cm}$$

2. Ans. (A,B,C)



Sol.

$$\mathbf{J}_N = \mathbf{J}$$

$$\mathbf{J}_f = m\mathbf{v}_{cm}$$

$$J \frac{4R}{5} - J_f R = \frac{2}{5} m R^2 \omega$$

$$\frac{4}{5}J - J_f = \frac{2}{5}mv_{cm} = \frac{2}{5}J_f$$

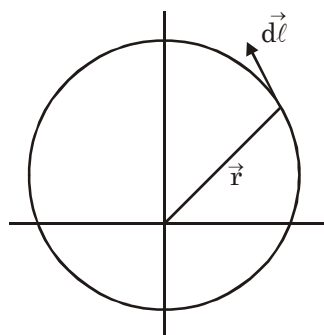
$$\Rightarrow \frac{4}{5}J = \frac{7}{5}J_f \Rightarrow J_f = \frac{4}{7}J$$

$$\mu J > \frac{4}{7} J \Rightarrow \mu > \frac{4}{7} \text{ for rolling}$$

3. Ans. (A,B)

Sol. $\vec{E} = \frac{\vec{F}}{q} \Rightarrow (A)$

Assume O is at origin, consider a circular loop of radius r , $\vec{r}$ will be perpendicular to $\vec{d\ell}$



As $\vec{E}$ is along $\vec{r}$

$$\text{So } \int \vec{E} \cdot \vec{d\ell} = 0$$

Consider sphere of radius r , $\vec{E}$ will be along $\vec{ds}$ at any position

$$\oint \vec{E} \cdot d\vec{s} = \oint (E)(ds)$$

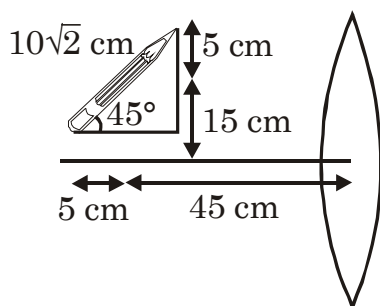
$$= E \oint ds = \frac{Q(1-\sqrt{\alpha r})}{4\pi\epsilon_0 r^2} (4\pi r^2) = \frac{Q}{\epsilon_0} (1-\sqrt{\alpha r})$$

4. **Ans. (B,C,D)**

Sol. Length of pencil is $10\sqrt{2}$ cm

Position of (A) $[-(45 + 5), (15 - 5)]$

$$\Rightarrow (-50, 10)$$



$$\frac{1}{v} - \frac{1}{(-50)} = \frac{1}{20}$$

$$v = 33.3 \text{ cm}$$

$$\text{magnification } \frac{h_1}{h_0} = \frac{-v}{u}$$

$$\Rightarrow h_1 = -h_0 \frac{v}{u} = 10 \frac{100/3}{50} = 6.6 \text{ cm}$$

Position of B (40 cm, 20 cm)

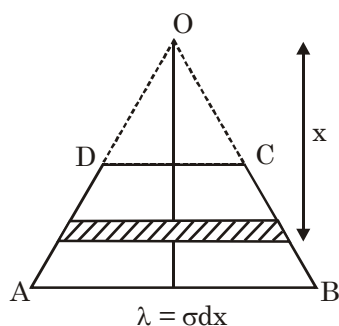
$$\frac{1}{v} - \frac{1}{-40} = \frac{1}{20} \Rightarrow v = 40 \text{ cm}$$

$$\text{magnification } h_1 = 20 \times \frac{40}{40} = 20 \text{ cm}$$

Position of C (36 cm, 12 cm)

$$\text{Magnification } h_1 = 15 \frac{36}{45} = 12 \text{ cm}$$

5. Ans. (B,C)



Sol.

$$dE = 2 \left(\frac{1}{4\pi\epsilon_0} \right) \frac{\sigma dx}{x} \left(\frac{1}{2} \right)$$

$$dE = \frac{\sigma}{4\pi\epsilon_0} \frac{dx}{x}$$

$$E = \frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$$

6. Ans. (A,C)

Sol. $\omega R = u = \text{constant}$

$$\omega \frac{dR}{dt} + R \frac{d\omega}{dt} = 0$$

$$\therefore \frac{d\omega}{dt} = -\frac{\omega}{R} \frac{dR}{dt}$$

In time dt radius decreases by dR

$$-2\pi R dR = u dt$$

$$\therefore \frac{dR}{dt} = -\frac{u}{2\pi R}$$

$$\alpha = \frac{u^2 d}{2\pi R^3}$$

$$\tau = \frac{I u^2 d}{2\pi R^3}$$

$$\therefore F = \frac{I u^2 d}{2\pi R^4}$$

7. Ans. (A,B,C,D)

Sol. $\vec{E} = (10 - 5x)\hat{i}$

$$m = 5 \text{ kg}$$

$$\vec{F} = (10 - 5x)\hat{i}$$

$$(A) F \propto -x$$

So motion of charge particle is oscillatory

$$(b) F = 0 \text{ at } x = 2$$

Oscillation about $x = 2$

So maximum displacement is 4

$$F = 10 - 5x$$

$$a = \frac{10 - 5x}{2}$$

$$a = 2 - x$$

$$a = -(x - 2)$$

$$\text{so } \omega^2 = 1$$

$$\omega = 1$$

(C) $v_{\max} = A\omega$

$A = 2$

$\omega = 1$

$v_{\max} = 2$

(D) velocity at mean position at $x = 2$

8. **Ans. (D)**

Sol. $\frac{3}{2v_1} + \frac{1}{45} = \frac{1}{2R} \quad \dots(i)$

$-\frac{3}{2v_1} = \frac{1}{6R} \quad \dots(ii)$

from (i) and (ii)

$R = 30 \text{ cm}$

9. **Ans. (D)**

Sol. $S_y = u_y t + \frac{1}{2} a_y t^2$

$0.01 = 0 + \frac{1}{2} \left(\frac{qE}{m} \right) t^2$

$t = \frac{10 \text{ cm}}{3 \text{ m/s}} = \frac{0.1}{3} \text{ sec}$

$0.01 = \frac{1}{2} \left(\frac{qE}{m} \right) \left(\frac{0.1}{3} \right)^2$

$E = 112500 \text{ N/c}$

$\Delta V = Ed = 112500 \times 0.01 \text{ Volt}$
 $= 1125 \text{ Volt.}$

10. **Ans. (A)**

Sol. $v_x = 3 \text{ m/s}$

$v_y = u_y a_y t = 0 + \left(\frac{qE}{m} \right) t$

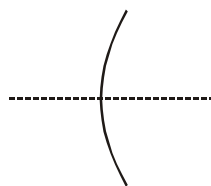
$= \left(\frac{100 \times 1.6 \times 112500 \times 10^{-19}}{10^{-13}} \right) \left(\frac{0.1}{3} \right)$

$= 0.6 \text{ m/s}$

11. **Ans. (A,B,D)**

Sol. $\frac{1}{v} - \frac{1.5}{u} = \frac{1-1.5}{R}$

$\frac{1}{v} = -\frac{1}{2R} + \frac{35}{2u}$



$RO \Rightarrow u = -ve$

$\therefore v$ is also $-ve$

Real object virtual image only.

No real object real image.

virtual object $u = +ve$

v can be positive and negative

virtual object real image and virtual object, virtual image possible.

12. **Ans. (B,C,D)**

Sol. $\frac{1.5}{v} = -\frac{1}{u} = \frac{1.5-1}{R}$

$\frac{3}{2v} = -\frac{1}{u} + \frac{1}{2R}$

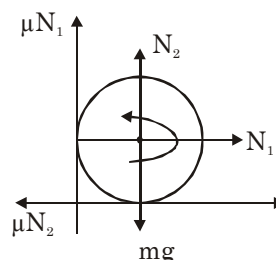
$\frac{1}{v} = \frac{2}{3} \left(\frac{1}{u} + \frac{1}{2R} \right)$

SECTION-II

1. **Ans. 5.00**

Sol. $\mu N_1 + N_2 = mg \quad \dots(i)$

$N_1 = \mu N_2 \quad \dots(ii)$



$N_2 = \frac{mg}{(1 + \mu^2)}$

$N_1 = \frac{\mu mg}{(1 + \mu^2)}$

Torque about center of mass

$$\mu(N_1 + N_2)R = \frac{1}{2}mR^2\alpha$$

$$\alpha = \frac{2\mu(N_1 + N_2)}{mR} = \frac{2\mu(1 + \mu)g}{(1 + \mu^2)R}$$

$$\omega^2 = \omega_0^2 - 2\alpha\theta$$

$$\theta = \frac{\omega_0^2}{2\alpha}$$

2. Ans. 1.00

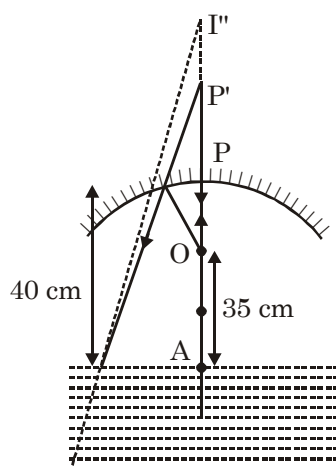
Sol. For the reflection at the concave mirror,

$$u = -10 \text{ cm}; v = ?; f = -15 \text{ cm}$$

From the mirror formula, we have

$$v = \frac{uf}{u - f} = \frac{(-10) \times (-15)}{-10 + 15} = \frac{150}{5} = +30 \text{ cm}$$

The positive sign indicates that the image is formed on the other side of the concave mirror.



Now, the image formed by the concave mirror serves as virtual object for refraction at water surface which takes place from air to water. So,

$$\mu = \frac{\text{Apparent height}}{\text{Real height}}$$

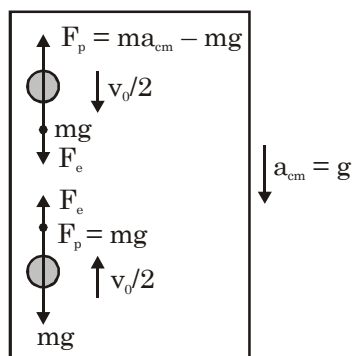
$$\therefore AI'' = \text{Apparent height} = \mu \times \text{real height} \\ = \frac{4}{3} \times 75 = 100 \text{ cm}$$

3. Ans. 13.33

Sol. From equilibrium of Q_1

$$\left[\frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{h_0^2} = mg \right]$$

in COM frame



Resultant of mg and pseudo force is zero on each mass.

At closed distance velocity of each is zero in COM frame. Let closest separation be x

Loss in KE

$$2 \frac{1}{2} m \left(\frac{v_0}{2} \right)^2 = \frac{1}{4\pi\epsilon_0} Q_1 Q_2 \left(\frac{1}{h_0} - \frac{1}{x} \right)$$

(Gain in electrostatic PE)

On solving $x = 13.33 \text{ cm}$

4. Ans. 9.00

Sol. To form image at infinity object should be at the focal length of lens.

$$S = t \left(1 - \frac{1}{\mu} \right) \text{ for slab}$$

$$\therefore d = 25 - S = f$$

$$25 - t \left(1 - \frac{1}{\mu} \right) = 22$$

$$t \left(1 - \frac{1}{3/2} \right) = 3$$

$$\frac{t}{3} = 3$$

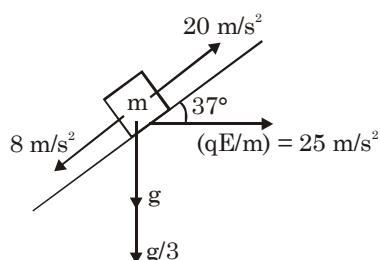
$$t = 9 \text{ cm}$$

$\therefore$ thickness of slab should be 9cm.

5. Ans. 1.00

Sol. FBD of block

$$a_{\text{net}} = 12 \text{ m/s}^2, \text{ up the incline plane}$$



$$S = ut + \frac{1}{2}gt^2$$

$$3.6 \times \frac{5}{3} = \frac{1}{2} \times 12 \times t^2$$

$$\Rightarrow t = 1 \text{ sec}$$

6. Ans. 4.00

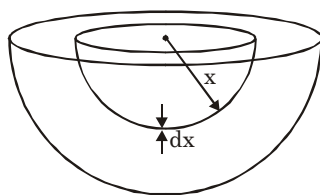
$$\text{Sol. } 1.25 = \frac{1}{2}(10)t^2$$

$$\therefore t = 0.5 \text{ sec}$$

$$\text{Distance of the cylinder from the truck} \\ = 8 \times 0.5 = 4\text{m}$$

7. Ans. 3.00

$$\text{Sol. } V = \int_a^b \frac{K(2\pi x^2 dx)\rho}{x}$$



$$V = 2\pi k\rho \left[\frac{b^2 - a^2}{2} \right] = \pi\rho k(b^2 - a^2)$$

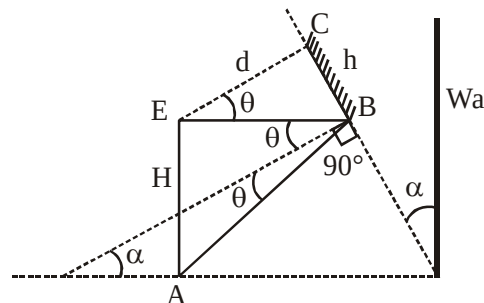
$$b = 2a$$

$$v = 3\pi k\rho b^2$$

8. Ans. 0.37 to 0.38

$$\text{Sol. } \frac{h}{d} = \tan \theta$$

Using triangle EAB



$$\frac{\frac{d}{\cos \theta}}{\sin \{90 - (\alpha + \theta)\}} = \frac{H}{\sin 2\theta}$$

$$\Rightarrow \frac{d}{\cos \theta(\alpha + \theta)} = \frac{H}{2\sin \theta}$$

$$\Rightarrow 2d = H[\cos \alpha \cot \theta - \sin \alpha]$$

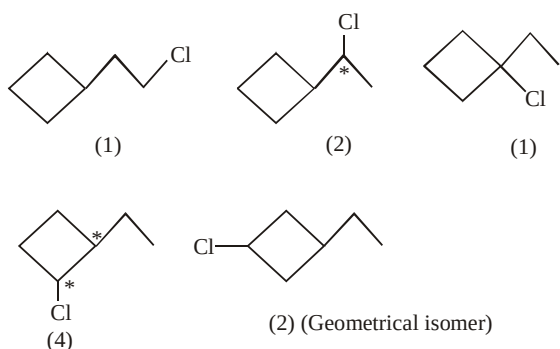
$$= H \cos \alpha \left(\frac{d}{h} \right) - H \sin \alpha$$

$$\Rightarrow h = \frac{Hd \cos \alpha}{2d + H \sin \alpha}$$

$$= \frac{\frac{3}{2} \times 3 \times \frac{3}{5}}{2 \times 3 + \frac{3}{2} \times \frac{4}{5}} = \frac{27}{36 \times 2} = \frac{3}{8}$$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans.(A,C,D)
2. Ans.(A,D)
3. Ans.(A,C,D)
4. Ans.(A,B,C,D)
5. Ans.(A,B,D)
6. Ans. (A,C)
7. Ans. (B,D)
8. Ans. (B,C)
9. Ans (C)



10. Ans (A)
11. Ans. (B)
12. Ans. (D)

SECTION-II

1. Ans(2.00)

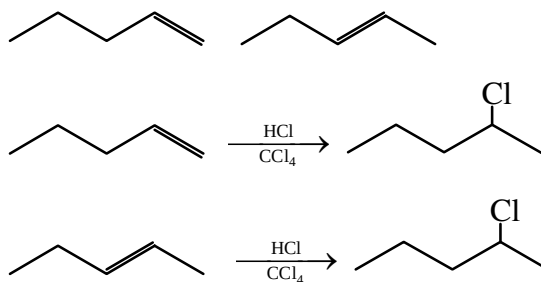
$$\log \frac{nR}{V} = 0.3010 = \log 2$$

$$n = \frac{2 \times 82.1 \times 10^{-3}}{0.0821} = 2$$

2. Ans (0.25)
3. Ans.(0.14)

$$[\text{Given : } \ln\left(\frac{1}{0.14}\right) =$$

4. Ans. (4.00)
5. Ans. (7.00)
6. Ans (2.00)



7. Ans.(5.90 or 5.92)
8. Ans.(5.00)

PART-3 : MATHEMATICS**SOLUTION****SECTION-I**

1. Ans.(A,B,D)
Sol. Sum of given number = $1 + 2 + 3 + \dots + 9 = 45$.
sum of 7 digits must be a multiple of 9.
 $\therefore$ pair of left numbers are
 $= \{(1,8), (2,7), (3,6), (4,5)\}$
corresponding to each pair the number of
seven digit number = $\underline{7}$
 $\therefore Q|n| = 4.\underline{7}$
2. Ans.(A,B,C)
Sol. Coefficient of x^{324} is
 $= 1^2 - 2^2 + 3^2 - 4^2 + \dots + 24^2 + 25^2$
 $= 325$
 $\therefore \sum_{k=0}^8 {}^8C_k = 2^8 = 256, \sum_{s=0}^4 {}^4C_s = 2^4 = 16, \sum_{r=0}^2 {}^2C_r = 4$
 $\therefore$ option(B) = $256 + 4 \times 16 + 4 + 1 = 325$.

3. Ans.(A,B)

Sol. $(\cos 3\theta + \sin 3\theta) + (2\sin 2\theta - 3) \cdot (\sin \theta - \cos \theta)$
 $\Rightarrow (4 \cos^3 \theta - 3 \cos \theta + 3 \sin \theta - 4 \sin^3 \theta) + (4 \sin \theta \cos \theta - 3) (\sin \theta - \cos \theta)$
 $\Rightarrow 3(\sin \theta - \cos \theta) - 4(\sin \theta - \cos \theta) (1 + \sin \theta \cos \theta) + (4 \sin \theta \cos \theta - 3) \cdot (\sin \theta - \cos \theta)$
 $= (\sin \theta - \cos \theta) (3 - 4 - 4 \sin \theta \cos \theta + 4 \sin \theta \cos \theta - 3)$
 $= -4(\sin \theta - \cos \theta)$
 $\Rightarrow (\sin \theta - \cos \theta) < 0$
 $\sin\left(\theta - \frac{\pi}{4}\right) < 0$
 $\Rightarrow (2n-1)\pi < \theta - \frac{\pi}{4} < 2n\pi$
 $\Rightarrow 2n\pi - \frac{3\pi}{4} < \theta < 2n\pi + \frac{\pi}{4}$

ALLEN**4. Ans. (B,C)****Sol.** Adding $\sin^2 B \sin^2 C$ to both sides

$$3\sin^2 B \sin^2 C = (\sin^2 B + \sin^2 C - \sin^2 A)^2$$

$$\Rightarrow 3b^2 c^2 = (b^2 + c^2 - a^2)^2$$

$$\Rightarrow \left(\frac{b^2 + c^2 - a^2}{2bc} \right)^2 = \frac{3}{4}$$

$$\Rightarrow \cos^2 A = \frac{3}{4}$$

$$\Rightarrow \cos A = \pm \frac{\sqrt{3}}{2}$$

$$\Rightarrow A = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

5. Ans. (A,B,C)

$$\text{Sol. } x = \log_2 \left[1 + \frac{1}{2} ({}^{12}C_1 + {}^{12}C_2 + {}^{12}C_3 + \dots + {}^{12}C_{11}) \right]$$

$$= \log_2 \left[1 + \frac{1}{2} (2^{12} - 2) \right] = \log_2 \left(\frac{2^{12} - 2 + 2}{2} \right)$$

$$= \log_2 2^{11} = 11$$

$$\log_y 3 = 4 \Rightarrow y^4 = 3 \text{ and}$$

$$\log_{y^2} 27 = \frac{3z}{2} \Rightarrow \log_{\sqrt{3}} 27 = \frac{3z}{2}$$

$$\Rightarrow \frac{3z}{2} = 6 \Rightarrow z = 4$$

$$\text{Now, } z^2 - y^4 = 16 - 3 = 13$$

$$x^2 - y^4 + z^2 = 13 + 121 = 134$$

$$x + z^2 \div y^4 = 11 + \frac{16}{3} = \frac{49}{3}$$

6. Ans. (D)**Sol.** Slope of line = a

$$\text{x-intercept} = -\frac{b}{a}$$

$$\text{y-intercept of line} = b$$

$$\text{When } a > 0 \text{ and } b < 0$$

$$\Rightarrow \text{x-intercept} + \text{ve}$$

$$\text{given conic } \frac{x^2}{a} - \frac{y^2}{b} = 1$$

$$\frac{x^2}{|a|} + \frac{y^2}{|b|} = 1 \Rightarrow \text{an ellipse}$$

7. Ans. (A,B,D)**Sol.** Equations are $(ax - b)(x - 1) = 0$

$$\& (bx - c)(x + 1) = 0$$

$$\Rightarrow \text{Roots are } \frac{b}{a}, 1 \text{ and } \frac{c}{b}, -1$$

$$\text{Case-I: } \frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$$

$$\text{Case-II: } \frac{b}{a} = -1 \& \frac{c}{b} \neq 1$$

$$\Rightarrow -a = b \neq c$$

$$\text{Case-III: } \frac{c}{b} = 1 \& \frac{b}{a} \neq -1 \Rightarrow b = c \neq -a$$

8. Ans. (A,B,C,D)

$$\text{Sol. } 4 \cos^2 x = |\cos x| (3 - 4 \cos^2 x)$$

$$\text{If } \cos x > 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3} \text{ or } \frac{5\pi}{3}$$

$$\text{If } \cos x < 0 \Rightarrow \cos x = -\frac{1}{2} \Rightarrow x = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3}$$

Paragraph for Question 9 & 10**9. Ans. (A)****Sol.** Equation of OQ is $y = mx$

$$m^2 x^2 = 4x - 20$$

$$\Rightarrow m^2 x^2 - 4x + 20 = 0$$

$$D = 0$$

$$\Rightarrow 16 - 4m^2 \cdot 20 = 0 \Rightarrow 1 - 5m^2 = 0$$

$$\Rightarrow m = \frac{1}{\sqrt{5}} \therefore y = \frac{1}{\sqrt{5}} x$$

$$\Rightarrow \frac{1}{5} x^2 = 4x - 20$$

$$\Rightarrow x^2 - 20x + 100 = 0$$

$$x = 10$$

$$y^2 = 8$$

$$\Rightarrow y = 2\sqrt{2} \quad Q(10, 2\sqrt{2})$$

10. Ans. (C)**Sol.** Area of ΔOPQ is**Paragraph for Question 11 & 12****11. Ans. (B)**

$$\text{Sol. } \Delta_1 = \frac{1}{2} R^2 (2A), \Delta_2 = \frac{1}{2} R^2 (2B), \Delta_3 = \frac{1}{2} R^2 (2C)$$

$$\Delta_1 : \Delta_2 : \Delta_3 = A : B : C = 2 : 3 : 4$$

$$2\lambda + 3\lambda + 4\lambda = \pi \Rightarrow \lambda = \frac{\pi}{9}$$

$$PL : PM : PN = \cos \frac{2\pi}{9} : \cos \frac{3\pi}{9} : \cos \frac{4\pi}{9}$$

12. Ans. (B)

Sol. Area of $\Delta LMN = \Delta PLM + \Delta PKL + \Delta PKM$

$$= \frac{1}{2}(PL)(PM)\sin(180^\circ - A)$$

$$+ \frac{1}{2}PK \cdot PL \sin(180^\circ - B)$$

$$= \frac{1}{2} \times (R \cos B)(R \cos C) \sin A$$

$$+ \frac{1}{2}R \cos C \cos A \sin B$$

$$+ \frac{1}{2}(R \cos A)(R \cos B) \sin C$$

$$= \frac{R^2}{2}(\sin A \cos B \cos C + \cos A \cos B \sin B$$

$$+ \cos A \cos B \cos C)$$

$$= \frac{R^2(\cos A \cos B \cos C)(\tan A + \tan B + \tan C)}{2}$$

$$= \frac{R^2}{2} \times \sin A \sin B \sin C$$

$$\Rightarrow \text{Area of } \Delta LMN = \frac{\Delta}{4}$$

SECTION-II

1. Ans. 3.50

Sol. Point of parabola $y^2 = 2x$ is $\left(\frac{t^2}{2}, t\right)$

Normal at t is $y + tx - 2at - at^3 = 0$

$$y + tx - t - \frac{t^3}{2} = 0$$

But normal is given by

$$y - \sqrt{2}x - 14k\sqrt{2} = 0$$

$$\Rightarrow t = -\sqrt{2}$$

$$k = \frac{1}{2}$$

Point is $(1, -\sqrt{2})$

other end of normal chord is $(4, 2\sqrt{2})$

Length of normal chord

$$\sqrt{9+18} = \sqrt{27} = 3\sqrt{3}$$

$$a = 3, k = \frac{1}{2}$$

2. Ans. 0.50

Sol. Let $y = mx + c$ is tangent to $x^2 + y^2 = 4$ and

$$\frac{x^2}{5} + \frac{y^2}{3} = 1$$

$$c^2 = 4(1+m^2) = 5m^2 + 3$$

$$\Rightarrow m^2 = \frac{4-3}{5-4} = 1$$

$$\Rightarrow m^2 = 1 \text{ \& } c^2 = 8$$

x-coordinate of point of tangent on circle is

$$(1+m^2)x^2 + 2mcx + c^2 - 4 = 0$$

$$\Rightarrow (2)x^2 - 2cx + c^2 - 4 = 0$$

$$\Rightarrow 2x^2 - 2cx + 4 = 0 \Rightarrow x^2 - cx + 2 = 0$$

$$\Rightarrow x^2 - 2\sqrt{2}x + 2 = 0 \Rightarrow x = \sqrt{2}$$

$$\therefore A(\sqrt{2}, \sqrt{2})$$

Point of tangent of ellipse

$$B\left(-\frac{5}{2\sqrt{2}}, -\frac{5}{2\sqrt{2}} + 2\sqrt{2}\right) = \left(-\frac{5}{2\sqrt{2}}, \frac{3}{2\sqrt{2}}\right)$$

Length of common tangent

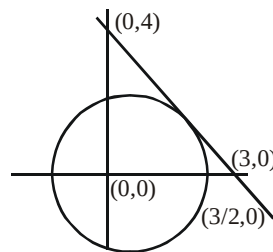
$$AB = \frac{1}{2} = \frac{1}{L}$$

3. Ans. 1.66 or 1.67

Sol. Let equation circle is

$$\left(x - \frac{3}{2}\right)^2 + (y-2)^2 + \lambda\left(\frac{x}{3} + \frac{y}{4} - 1\right) = 0$$

it passes through $\left(\frac{3}{2}, 0\right)$



$$\lambda = \frac{4}{2(3)} = \frac{2}{3}$$

equation of circle will be

$$x^2 + y^2 - \frac{x}{3} - 2y - \frac{7a}{4} = 0$$

$$\text{radius} = \sqrt{\frac{1}{36} + 1 + \frac{7}{4}} = \sqrt{\frac{1+36+63}{36}} = \sqrt{\frac{100}{36}}$$

$$= \frac{10}{6} = \frac{5}{3} \Rightarrow 3r = 5$$

4. **Ans. 1.11**

Sol. Equation of chord is

$$\frac{x-5}{\cos \alpha} = \frac{y-3}{\sin \alpha} = r$$

Point $(5 + r \cos \alpha, 3 + r \sin \alpha)$ of chord is one of its end point

$$2(5 + r \cos \alpha)^2 - 3(3 + r \sin \alpha)^2 = 6$$

$(5, 3)$ is midpoint of chord iff sum of roots = 0

$$20 \cos \alpha - 18 \sin \alpha = 0$$

$$\frac{\cos \alpha}{9} = \frac{\sin \alpha}{10} \Rightarrow \tan \alpha = \frac{10}{9} = m$$

5. **Ans. 2.00**

Sol.
$$\frac{a \cos B + b \cos A}{c \cos A + a \cos C} = \frac{1}{\sqrt{3}}$$

$$\frac{c}{b} = \frac{1}{\sqrt{3}} \quad \dots(1)$$

$$\& (A + C) + B = 180^\circ \Rightarrow B = 60^\circ$$

from equation (1)

$$\frac{\sin C}{\sin B} = \frac{1}{\sqrt{3}} \Rightarrow \sin C = \frac{1}{2} \Rightarrow C = 30^\circ$$

$$\& A = 90^\circ$$

$$\frac{3 \left(1 + \frac{3}{4} + \frac{1}{4} \right)}{\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right)^2} = \frac{3(2)}{3} = 2$$

6. **Ans. 4.00**

Sol. **Case-I :** 4 4 4 0 $\rightarrow$ 3 ways
(0 can't be in first position)

$$\text{Case-II : } 4 \ 4 \ 2 \ 2 \rightarrow \frac{4!}{2! \ 2!} = 6$$

$$\text{Case-III : } 4 \ 4 \ 3 \ 1 \rightarrow \frac{4!}{2!} = 12$$

$$\text{Case-IV : } 4 \ 3 \ 3 \ 2 \rightarrow \frac{4!}{2!} = 12$$

$$\text{Case-V : } 3 \ 3 \ 3 \ 3 \rightarrow 1$$

$$\text{Total} = 34$$

7. **Ans. 9.00**

Sol.
$$\frac{r}{(r+1)} \cdot {}^{18}C_r = \left(1 - \frac{1}{r+1} \right) \cdot {}^{18}C_r$$

$$= {}^{18}C_r - \frac{18}{(r+1)} \cdot {}^{18}C_r = {}^{18}C_r - \frac{1}{19} {}^{19}C_{r+1}$$

$$\therefore T = \sum_{r=1}^{18} (-1)^{r-1} \cdot \frac{r}{(r+1)} \cdot {}^{18}C_r = {}^{18}C_1 - {}^{18}C_2 + {}^{18}C_3 - \dots$$

$$= {}^{18}C_0 - \frac{1}{19} ({}^{19}C_0 + {}^{19}C_1) = 1 + \frac{1}{19} - 1 = \frac{1}{19}$$

$$\Rightarrow \frac{1}{T} = 19$$

8. **Ans. 64.00**

Sol.
$$D = \begin{vmatrix} \alpha - \beta - \gamma & 2\alpha & 2\alpha \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$D = (\alpha + \beta + \gamma) \begin{vmatrix} 1 & 1 & 1 \\ 2\beta & \beta - \gamma - \alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma - \alpha - \beta \end{vmatrix}$$

$$C_3 \rightarrow C_3 - C_1 \ \& \ C_2 \rightarrow C_2 - C_1$$

$$D = (\alpha + \beta + \gamma) \begin{vmatrix} 1 & 0 & 0 \\ 2\beta & -(\alpha + \beta + \gamma) & 0 \\ 2\gamma & 0 & -(\alpha + \beta + \gamma) \end{vmatrix}$$

$$D = (\alpha + \beta + \gamma)^3 = 64.$$